

## Questions

### 1. Write a Program

Bob fires a cannon horizontally off a hill. The following data is the height of the cannon ball above the ground at different times. The sensor used to measure the height of the cannon above the ground is buggy. Write a function that removes any data point that is greater than the previous data point. Fit a second order polynomial to this data. Plot the cleaned data as data points and the fitted line on the same graph and add a legend, axis labels and a title. Calculate the derivative of the data (the speed of the ball in the y direction) using the data from the line of best fit. On a second plot, plot the derivative as points. Publish the script as a pdf.

#### Data:

|            |    |      |      |      |     |      |      |      |      |      |      |      |      |      |      |      |      |
|------------|----|------|------|------|-----|------|------|------|------|------|------|------|------|------|------|------|------|
| Time (ms)  | 0  | 50   | 750  | 800  | 825 | 1000 | 1250 | 1500 | 1700 | 1900 | 2000 | 2250 | 2500 | 2600 | 2750 | 3000 | 3100 |
| Height (m) | 10 | 9.99 | 9.43 | 9.36 | 9.5 | 9.0  | 8.44 | 7.75 | 7.11 | 6.39 | 6.0  | 4.94 | 3.75 | 5.0  | 2.44 | 1.0  | 0.39 |

#### Extensions:

- Calculate the derivative of the data by creating an inline function using the coefficients from the polynomial fit
- Find the time when the cannon hits the ground using the roots function
- Instead of using the original time data to generate the fitted data, create a finer sampled time array
- Calculate the  $R^2$  of the fit

#### Code:

```
%q1
```

```
% Practice Test Script
```

```
% Time data converted from ms to s
```

```
time = [0, 50, 750, 800, 825, 1000, 1250, 1500, 1700, 1900, 2000, 2250, 2500, 2600, 2750, 3000, 3100] * 1e-3;
```

```
% Height in m
```

```
height = [10, 9.99, 9.43, 9.36, 9.5, 9.0, 8.44, 7.75, 7.11, 6.39, 6.0, 4.94, 3.75, 5.0, 2.44, 1.0, 0.39];
```